



Exercise Set 4.1 Solution

Question 06

Let $\mathbf{u} = (4, 1, 2, 3)$, $\mathbf{v} = (0, 3, 8, -2)$, and $\mathbf{w} = (3, 1, 2, 2)$. Evaluate each expression.

(c) $\| -2\mathbf{u} \| + 2\|\mathbf{u}\|$ (d) $\|3\mathbf{u} - 5\mathbf{v} + \mathbf{w}\|$ (f) $\left\| \frac{1}{\|\mathbf{w}\|} \mathbf{w} \right\|$

Solution:

$$(c) \quad \| -2\mathbf{u} \| + 2\|\mathbf{u}\| = \| -2(4, 1, 2, 3) \| + 2\|(4, 1, 2, 3)\| \\ = \sqrt{(-8)^2 + (-2)^2 + (-4)^2 + (-6)^2} + 2\sqrt{4^2 + 1^2 + 2^2 + 3^2}$$

$$(d) \quad \|3\mathbf{u} - 5\mathbf{v} + \mathbf{w}\| = \|3(4, 1, 2, 3) - 5(0, 3, 8, -2) + (3, 1, 2, 2)\| \\ = \|(12 + 0 + 3, 3 - 15 + 1, 6 - 40 + 2, 9 + 10 + 2)\| \\ = \|(15, -11, -32, 21)\| = \sqrt{15^2 + (-11)^2 + (-32)^2 + 21^2}$$

$$(f) \quad \left\| \frac{1}{\|\mathbf{w}\|} \mathbf{w} \right\| = \left\| \left(\frac{3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{2}{\sqrt{14}} \right) \right\| = \sqrt{\left(\frac{3}{\sqrt{14}} \right)^2 + \left(\frac{1}{\sqrt{14}} \right)^2 + \left(\frac{2}{\sqrt{14}} \right)^2 + \left(\frac{2}{\sqrt{14}} \right)^2} = 1$$

Question 07

Show that if v is a nonzero vector in $\mathbb{R}^n$, then $\frac{1}{\|v\|}v$ has Euclidean norm 1.

Solution: Let $v = (v_1, v_2, \dots, v_n)$, so $\frac{1}{\|v\|}v = \frac{1}{\sqrt{v_1^2 + v_2^2 + \dots + v_n^2}}(v_1, v_2, \dots, v_n)$, so the

$$\text{Euclidean norm is } \left\| \frac{1}{\|v\|}v \right\| = \sqrt{\left(\frac{v_1}{\sqrt{v_1^2 + v_2^2 + \dots + v_n^2}} \right)^2 + \left(\frac{v_2}{\sqrt{v_1^2 + v_2^2 + \dots + v_n^2}} \right)^2 + \dots + \left(\frac{v_n}{\sqrt{v_1^2 + v_2^2 + \dots + v_n^2}} \right)^2} \\ = \sqrt{\left(\frac{v_1^2}{v_1^2 + v_2^2 + \dots + v_n^2} \right) + \left(\frac{v_2^2}{v_1^2 + v_2^2 + \dots + v_n^2} \right) + \dots + \left(\frac{v_n^2}{v_1^2 + v_2^2 + \dots + v_n^2} \right)} = \sqrt{\frac{v_1^2 + v_2^2 + \dots + v_n^2}{v_1^2 + v_2^2 + \dots + v_n^2}} = \sqrt{1}$$

Question 08

Let $v = (-2, 3, 0, 6)$. Find all scalars k such that $\|kv\| = 5$.

Solution: $\|kv\| = 5$

$$\|(-2k, 3k, 0, 6k)\| = 5; \quad \sqrt{(-2k)^2 + (3k)^2 + 0^2 + (6k)^2} = 5; \\ \sqrt{4k^2 + 9k^2 + 36k^2} = 5; \quad 49k^2 = 25; \quad k = \pm \frac{5}{7}$$

Question 10

(a) Find two vectors in $\mathbb{R}^2$ with Euclidean norm 1 whose Euclidean inner product with $(3, -1)$ is zero.

Solution: Let $u = (3, -1)$ and $v = (\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}})$. So, $u \cdot v = 3 * \frac{1}{\sqrt{10}} + (-1) * \frac{3}{\sqrt{10}} = 0$

(b) Show that there are infinitely many vectors in with Euclidean norm 1 whose Euclidean inner product with $(1, -3, 5)$ is zero.

Solution: Let $v = (\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}, 0)$, $w = (\frac{-5}{\sqrt{26}}, 0, \frac{1}{\sqrt{26}})$, $v = (\frac{5}{\sqrt{26}}, 0, \frac{-1}{\sqrt{26}})$

Question 20

Solution: $u \cdot v = \frac{1}{4}\|u + v\|^2 - \frac{1}{4}\|u - v\|^2 = \frac{1}{4}(1)^2 - \frac{1}{4}(5)^2$

Question 21

Use Theorem 4.1.6 to show that u and v are orthogonal vectors in $\mathbb{R}^n$ if $\|u + v\| = \|u - v\|$. Interpret this result geometrically in $\mathbb{R}^2$.

Solution: Theorem 4.1.6 $u \cdot v = \frac{1}{4}\|u + v\|^2 - \frac{1}{4}\|u - v\|^2$. Now substitute $\|u + v\| = \|u - v\|$, we have $u \cdot v = \frac{1}{4}\|u - v\|^2 - \frac{1}{4}\|u - v\|^2$, so $u \cdot v = 0$. Hence it is orthogonal.

Question 24

Prove the following generalization of Theorem 4.1.7. If $v_1, v_2, \dots, v_n$ are pairwise orthogonal vectors in $\mathbb{R}^2$, then

$$\|v_1 + v_2 + \dots + v_r\|^2 = \|v_1\|^2 + \|v_2\|^2 + \dots + \|v_r\|^2$$

Solution: Proof

$$\begin{aligned}\|v_1 + v_2 + \dots + v_r\|^2 &= \|v_1 + (v_2 + \dots + v_r)\|^2 \\ &= \|v_1\|^2 + \|v_2 + v_3 + \dots + v_r\|^2\end{aligned}$$

We know $\|v_1 + v_2\|^2 = \|v_1\|^2 + \|v_2\|^2$ $\because v_i v_j$ are orthogonal

Apply this procedure r-1 times

$$\begin{aligned}&\vdots \\ &\vdots \\ &= \|v_1\|^2 + \|v_2\|^2 + \dots + \|v_r\|^2\end{aligned}$$

Question 25

Prove: If u and v are $n \times 1$ matrices and A is $n \times n$ an matrix, then

$$(v^T A^T A u)^2 \leq (u^T A^T A u)(v^T A^T A v)$$

Solution: Let $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ be vectors and $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ in $\mathbb{R}^n$ and let A be an $n \times n$ matrix.

So, letting $u = Au$ and $v = Av$ and this multiplication possible. Hence the inner product is $(a, b) = a \cdot b = b^T a$.

$$(u, v) = Au \cdot Av = (Av)^T Au$$

So we have

$$(u, v) = v^T A^T A u$$

In special cases we have

$$(u, u) = u^T A^T A u \text{ and } (v, v) = v^T A^T A v$$

The Cauchy-Schwarz inequality is

$$(u, v)^2 \leq (u, u)(v, v)$$

By substituting the previous we get

$$(u^T A^T A u)^2 \leq (u^T A^T A u)(v^T A^T A v)$$

Solution:

Question 26

Use the Cauchy-Schwarz inequality to prove that for all real values of a, b , and θ .

$$(a \cos \theta + b \sin \theta)^2 \leq a^2 + b^2$$

Solution: Take $u = (a, b)$ and $v = (\cos(\theta), \sin(\theta))$. Then,

$$\|u\|^2 = a^2 + b^2, \quad \|v\|^2 = \cos^2(\theta) + \sin^2(\theta) = 1, \quad \langle u, v \rangle = a \cos(\theta) + b \sin(\theta).$$

Hence, by Cauchy-Schwarz,

$$|\langle u, v \rangle|^2 \leq \|u\| \cdot \|v\| \implies (a \cos(\theta) + b \sin(\theta))^2 \leq a^2 + b^2$$

Question 27

Prove: If u, v , and w are vectors in R^k and k is any scalar, then

a) $u.(kv) = k(u.v)$

b) $u.(v + w) = u.v + u.w$

Solution: See solution manual.

Question 32

Prove parts (a), (b), and (c) of Theorem 4.1.5.

Solution: (a)

$$d(u, v) \geq 0, \implies d(u, v) = \|u - v\| = \sqrt{(u_1 - v_1)^2 + \dots + (u_n - v_n)^2}$$

Hence by property of squaring each component, it must greater and equal to zero.

(b)

$$d(u, u) = 0, \implies d(u, u) = \|u - u\| = \sqrt{(u_1 - u_1)^2 + \dots + (u_n - u_n)^2} = \|0\| = 0$$

Similar result for $d(v, v)$

(c) $d(u, v) = d(v, u)$

$$d(u, v) = \|u - v\| = \sqrt{(u_1 - v_1)^2 + \dots + (u_n - v_n)^2} = \sqrt{(-1(v_1 - u_1))^2 + \dots + (-1(v_n - u_n))^2}$$

$$\sqrt{(v_1 - u_1)^2 + \dots + (v_n - u_n)^2} = \|v - u\| = d(v, u)$$

Question 34

(a) Suppose that u and v are vectors in R^n . Show that

$$\|u + v\|^2 + \|u - v\|^2 = 2(\|u\|^2 + \|v\|^2)$$

(b) The result in part (a) states a theorem about parallelograms in R^2 . What is the theorem?

Solution: (a)

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 &= (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = \|\mathbf{u}\|^2 + 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2 \\ \|\mathbf{u} - \mathbf{v}\|^2 &= (\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}) = \|\mathbf{u}\|^2 - 2(\mathbf{u} \cdot \mathbf{v}) + \|\mathbf{v}\|^2\end{aligned}$$

Adding above two equations, we will have

$$\|u + v\|^2 + \|u - v\|^2 = (\|u\|^2 + 2(u \cdot v) + \|v\|^2) + (\|u\|^2 - 2(u \cdot v) + \|v\|^2) = 2(\|u\|^2 + \|v\|^2)$$

(b) *It states that the sum of the squares of the lengths of the four sides of a parallelogram equals the sum of the squares of the lengths of the two diagonals.*

Question 35

(a) If u and v are orthogonal vectors in R^n such that $\|u\| = 1$ and $\|v\| = 1$, then $d(u, v) =$.

(b) Draw a picture to illustrate this result

Solution: See solution manual.

Question 36

In the accompanying figure the vectors u, v , and $u - v$ form a triangle in R^2 , and θ denotes the angle between u and v . It follows from the law of cosines in trigonometry that $\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos\theta$. Do you think that this formula still holds if u and v are vectors in R^n ? Justify your answer.

Solution: Yes, it holds since dot product and norm of u and v ($u \cdot v, \|u\|\|v\|\cos\theta$) still valid in R^n .

Question 37

Indicate whether each statement is always true or sometimes false. Justify your answer by giving a logical argument or a counterexample.

Solution: See solution manual.